

SECURITY INCIDENT REPORT

Phase 09 | Brute-Force Authentication Investigation

Incident ID	SOC-HL-2026-0711	Severity	HIGH
Date	11 July 2026	Status	Closed — Lab Exercise
Target	WIN11-01 / 192.168.10.20	Attacker IP	192.168.10.30
Account	localadmin	Category	Credential Attack / Brute Force

Executive Summary

During a controlled SOC lab exercise, 192.168.10.30 repeatedly attempted to authenticate to the localadmin account on WIN11-01. Splunk recorded 31 failed logons (Windows Event ID 4625), followed by four successful network logons (Event ID 4624, Logon Type 3) from the same address. Taken together, the events show that the tested credentials were successfully used. The supplied logs do not show destructive activity, persistence, or data exfiltration after authentication.

KEY FINDING

Thirty-one failed attempts were followed by four successful localadmin network logons from the same source address.

Lab Architecture and Data Flow

Role	System	Address	Function
Attacker	Kali Linux	192.168.10.30	Nmap / NetExec activity in the isolated lab
Target	WIN11-01	192.168.10.20	Windows Security + Sysmon telemetry
SIEM	DC01 / Splunk	192.168.10.10	Receives logs on TCP 9997 and supports investigation

Scope and Method

- Reviewed Windows Security Event IDs 4625 and 4624 by source address, account, time, and logon type.
- Reviewed Sysmon Operational telemetry and validated Universal Forwarder collection into Splunk.
- Used filtered authentication results for the incident finding and retained broader searches as investigation context.

Detailed Timeline and Findings

Time	Event	Observation	Assessment
06:35–06:58	4625	31 failures from 192.168.10.30 against localadmin	Brute-force pattern
06:46:41	4624	Successful localadmin logon from 192.168.10.30; Type 3	Access confirmed
06:50:30	4624	Second successful network logon from same IP/account	Repeated access
06:58:40	4624	Third successful localadmin network logon	Credential reuse
10:34:55	4624	Fourth successful localadmin network logon	Later access
08:10:00	Sysmon 3	Network connection existed locally in Sysmon	Local logging confirmed
12:17:21	Sysmon 3	PowerShell 192.168.10.20 → 192.168.10.10:9997	Telemetry test
12:25:04	Sysmon 3	Second PowerShell connection to Splunk receiver	Forwarding validated

Confirmed Indicators

Indicator	Value	Interpretation
Attacker address	192.168.10.30	Source of 31 failed attempts and four successful logons
Target account	localadmin	Local administrative account exposed to network authentication
Target host	WIN11-01 / 192.168.10.20	System receiving the authentication activity
Logon type	3	Network logon; consistent with remote SMB/RPC authentication
Splunk receiver	192.168.10.10:9997	Validated telemetry destination; not an attacker connection

Analyst Assessment

The correlation is high confidence because the same source IP and account appear in both the failed and successful authentication records, and the successful records use Logon Type 3. Events labelled ANONYMOUS LOGON were excluded from the success determination. The four localadmin successes—not the anonymous events—support the compromise conclusion.

LIMITATION	The supplied logs confirm authentication success but do not prove command execution, persistence, privilege escalation, or data theft after access.
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Authentication Evidence

Figures 1–3 establish the volume, source, and targeted account for the failed authentication activity.

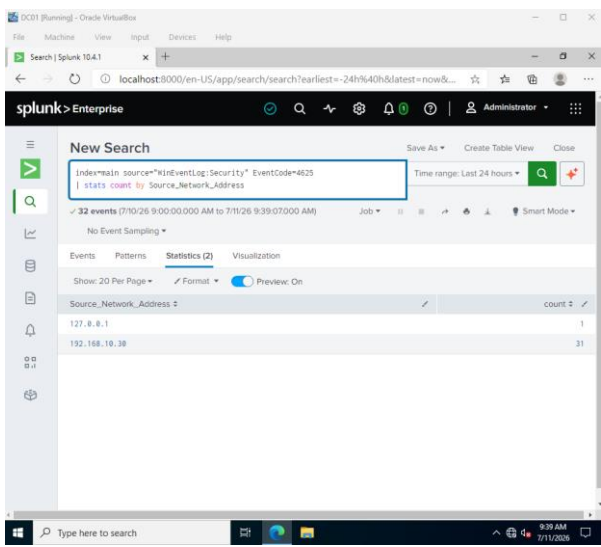


Figure 1. Failed logons grouped by source address.

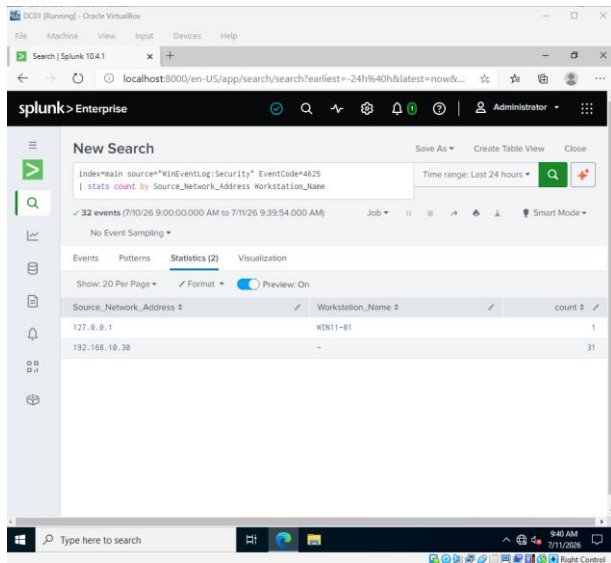


Figure 2. Source address and workstation field for failed logons.

INTERPRETATION

192.168.10.30 was responsible for 31 failed logons. The single 127.0.0.1 event is local loopback activity and is not attributed to the attacker.

Targeted Account

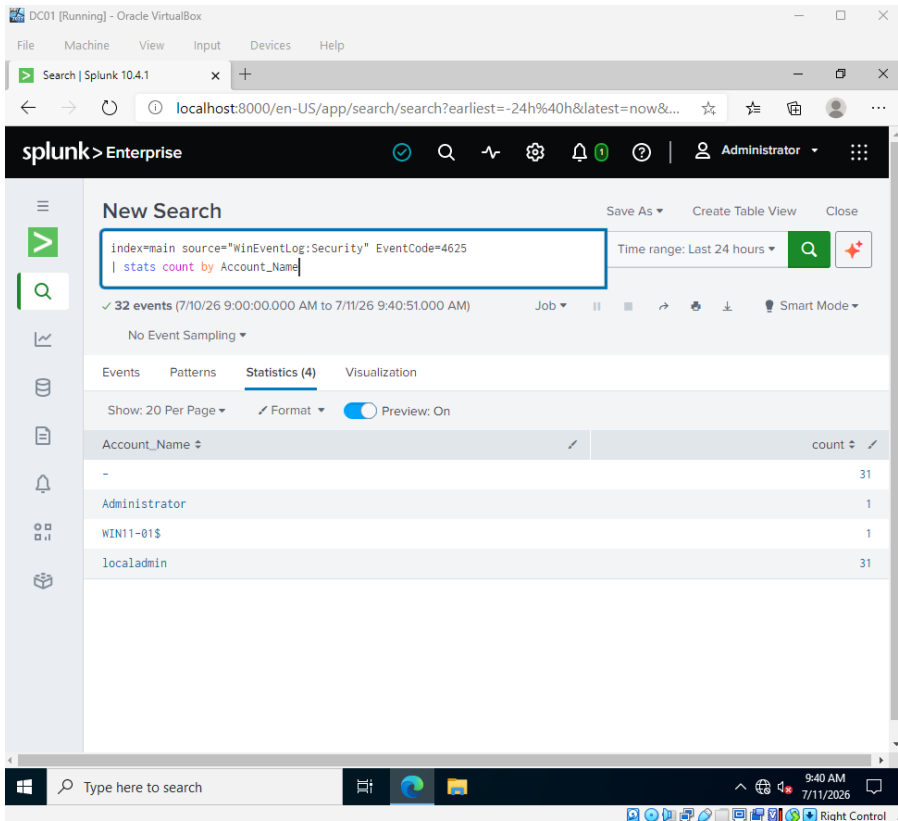


Figure 3. The localadmin account appears in 31 failed authentication events.

Timeline Construction — Initial Searches

These screenshots document the investigation process. They helped locate relevant time windows, but they are not used alone as proof because the searches were broad.

Broad 4624/4625 Timeline

The screenshot shows a Splunk search interface with the following search query:

```
index=main source="WinEventLog:Security"
(EventCode=4625 OR EventCode=4624)
| table _time EventCode Account_Name Source_Network_Address
| sort _time
```

The search results show 184 events from 7/10/26 9:00:00.000 AM to 7/11/26 9:41:20.000 AM. The table displays the following columns: time, EventCode, Account_Name, and Source_Network_Address. The results show a series of 4624 events from the SYSTEM account.

time	EventCode	Account_Name	Source_Network_Address
2026-07-10 09:00:09	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:01:34	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:04:02	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:06:49	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:07:05	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:07:49	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:08:48	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:09:19	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:10:43	4624	WIN11-01\$ SYSTEM	-
2026-07-10 09:11:16	4624	WIN11-01\$ SYSTEM	-

Figure 4. Broad authentication timeline used to establish the investigation window.

First Combined Attack Search

The screenshot shows a Splunk search interface with the following search query:

```
index=main
(source="Sysmon" OR source="WinEventLog:Security")
(Image="nmap" OR Image="hydra" OR EventCode=4625 OR EventCode=4624)
| table _time EventCode Image CommandLine Account_Name DestinationIp
| sort _time
```

The search results show 1548 events from before 7/11/26 10:13:18.000 AM. The table displays the following columns: time, EventCode, Image, CommandLine, Account_Name, and DestinationIp. The results show a series of 4624 events from the SYSTEM account, with some events showing the LOCAL SERVICE and NETWORK SERVICE accounts.

time	EventCode	Image	CommandLine	Account_Name	DestinationIp
2026-07-11 18:02:35	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ SYSTEM	
2026-07-11 18:02:34	4624			WIN11-01\$ LOCAL SERVICE	
2026-07-11 18:02:34	4624			WIN11-01\$ DWM-1	
2026-07-11 18:02:34	4624			WIN11-01\$ DWM-1	
2026-07-11 18:02:33	4624			WIN11-01\$ NETWORK SERVICE	

Figure 6. Exploratory combined search used while narrowing the event set.

ANALYST NOTE

This search was useful for exploration, but its OR logic also returned general 4624/4625 events. Hydra was not used in the exercise, and Nmap and NetExec ran on Kali, so their process images would not appear in Windows Sysmon.

Authentication Correlation and Final Proof

Correlation Review

New Search

```
Index=main
(source=Sysmon OR source=WinEventLog:Security)
(Image=nmap OR Image=hydra OR EventCode=4624)
| table _time EventCode Image CommandLine Account_Name DestinationIp
| sort _time
```

✓ 177 events (7/10/26 10:00:00 AM to 7/11/26 10:16:15 AM) No Event Sampling

_time	EventCode	Image	CommandLine	Account_Name	DestinationIp
2026-07-10 10:13:31	4624			WIN11-01\$ SYSTEM	
2026-07-10 10:14:51	4624			WIN11-01\$ SYSTEM	
2026-07-10 10:15:14	4624			WIN11-01\$ SYSTEM	
2026-07-10 10:15:38	4624			WIN11-01\$ Administrator	
2026-07-10 10:15:38	4624			WIN11-01\$ Administrator	
2026-07-10 10:15:46	4624			- WIN11-01\$	
2026-07-10 10:15:46	4624			- WIN11-01\$	
2026-07-10 10:18:15	4624			WIN11-01\$ SYSTEM	
2026-07-10 10:29:27	4624			WIN11-01\$ SYSTEM	

Figure 7. Exploratory Sysmon and authentication correlation view.

Successful Logons from the Attacker IP

New Search

```
Index=main source=WinEventLog:Security
EventCode=4624
Account_Name=localadmin
Source_Network_Address=192.168.10.30
| table _time EventCode Account_Name Source_Network_Address Logon_Type Workstation_Name
| sort _time
```

✓ 4 events (7/10/26 12:00:00 PM to 7/11/26 12:06:35 PM) No Event Sampling

_time	EventCode	Account_Name	Source_Network_Address	Logon_Type	Workstation_Name
2026-07-11 06:46:41	4624	localadmin	192.168.10.30	3	-
2026-07-11 06:58:30	4624	localadmin	192.168.10.30	3	-
2026-07-11 06:58:40	4624	localadmin	192.168.10.30	3	-
2026-07-11 10:34:55	4624	localadmin	192.168.10.30	3	-

Figure 8. Four successful localadmin network logons from 192.168.10.30.

FINAL PROOF

Figure 8 uses explicit EventCode=4624, Account_Name=localadmin, and Source_Network_Address=192.168.10.30 filters. It replaces the broad searches above.

Sysmon Event ID 3 — Diagnostic Sequence

The investigation initially returned zero Sysmon network-connection records in Splunk. Local validation then proved that Event ID 3 was enabled and present on WIN11-01.

Initial Splunk Result

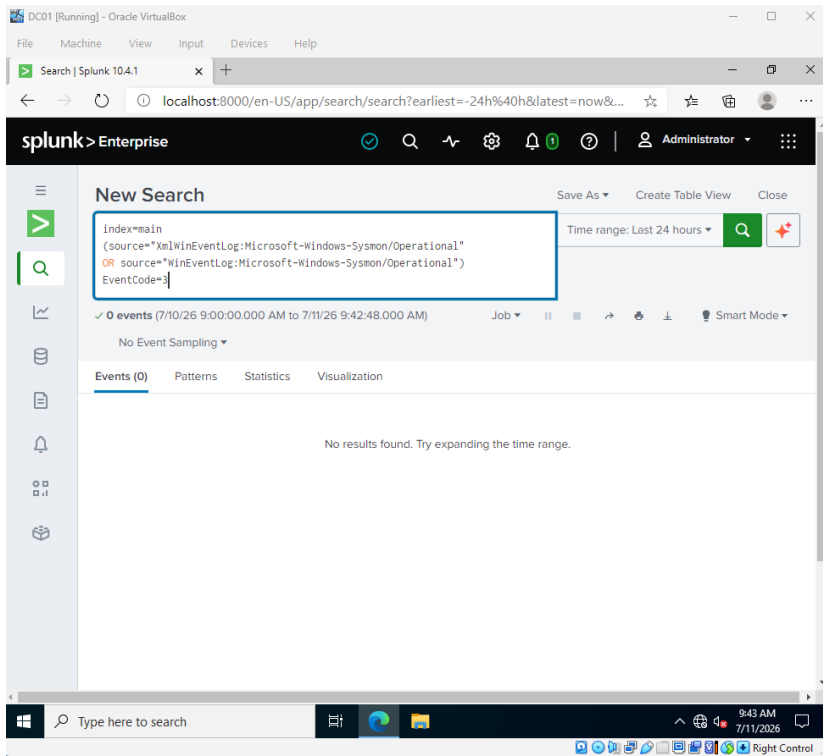


Figure 5. Event ID 3 initially returned no results in Splunk.

Local Windows Validation

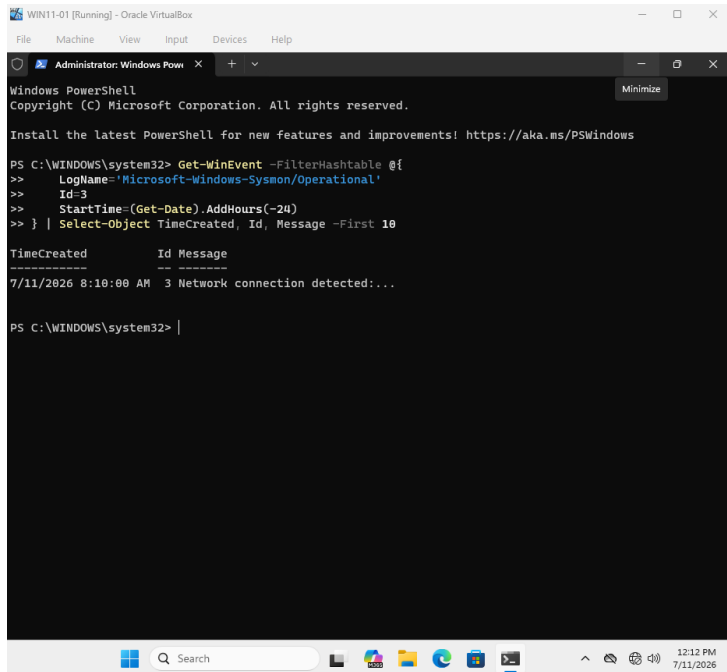


Figure 9. Local validation confirmed that Sysmon Event ID 3 was present on WIN11-01.

ROOT CAUSE BOUNDARY	Sysmon was functioning locally. The issue was therefore between collection/reload and Splunk ingestion, not the Sysmon NetworkConnect configuration itself.
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Forwarding Recovery and Telemetry Validation

The Universal Forwarder inputs.conf already contained the correct Sysmon channel and the service was running. Restarting SplunkForwarder reloaded the input, after which fresh Sysmon events arrived in Splunk.

Collection Restored

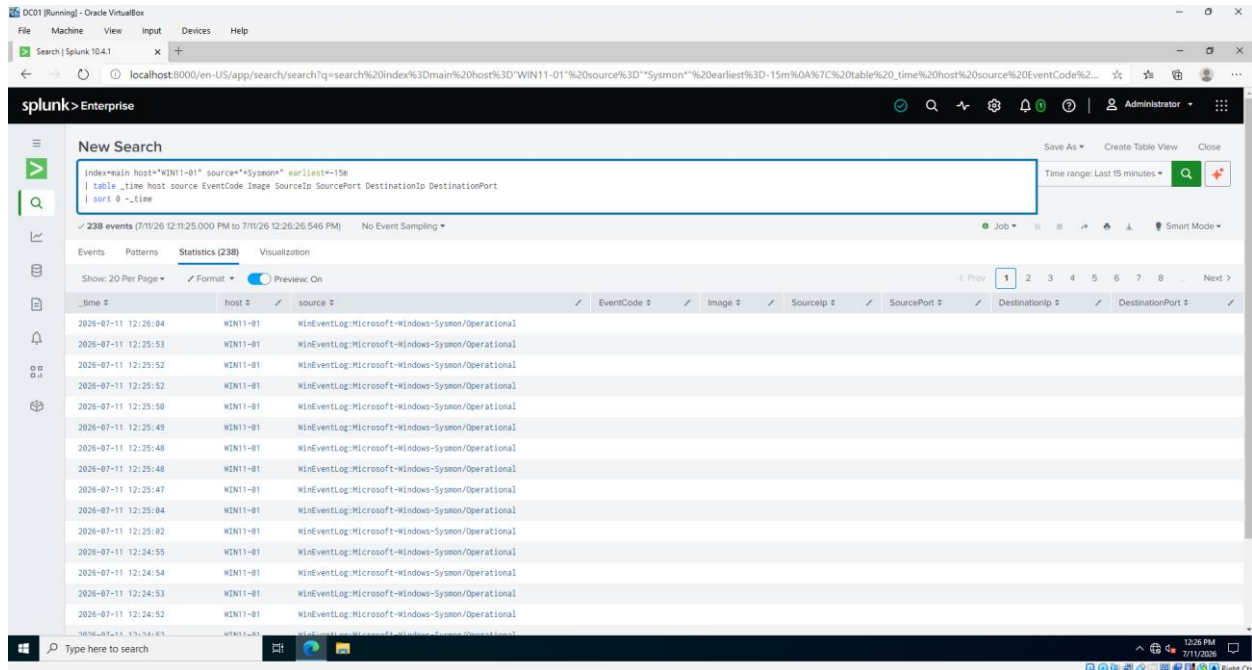


Figure 10 — 238 fresh events from the correct Sysmon source (RECOVERY VALIDATION).

Event ID 3 Extracted from XML

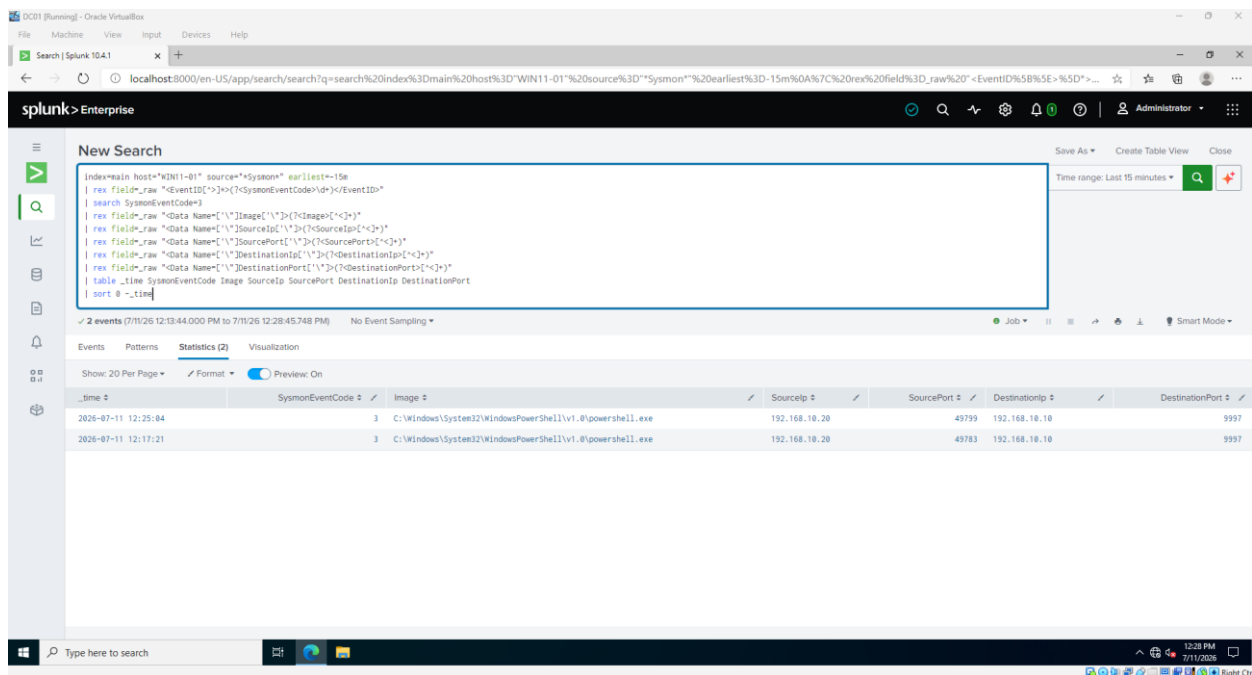


Figure 11 — Event ID 3 with process and network fields extracted (PRIMARY TELEMETRY EVIDENCE).

IMPORTANT

The Figure 11 connections are controlled PowerShell telemetry tests from 192.168.10.20 to Splunk port 9997. They verify visibility and are not attacker traffic.

Technical Analysis and Detection Logic

Root Cause

The Universal Forwarder inputs.conf already contained the correct Sysmon channel, and the SplunkForwarder service was running. Restarting the service reloaded the input and restored collection of fresh Sysmon events. Because renderXml=true was enabled, the events arrived in XML format and the investigation search extracted the required fields from _raw.

Authentication Search

Successful localadmin logons from the attacker IP

```
index=main source="WinEventLog:Security" EventCode=4624
Account_Name="localadmin" Source_Network_Address="192.168.10.30"
| table _time EventCode Account_Name Source_Network_Address Logon_Type
| sort 0 _time
```

Sysmon XML Extraction Search

Extract Event ID 3 network fields

```
index=main host="WIN11-01" source="*Sysmon*"
| rex field=_raw "<EventID[^>]*>(?!<SysmonEventCode>\d+)</EventID>"
| search SysmonEventCode=3
| rex field=_raw "<Data Name=[\"]Image[\"]>(?!<Image>[^\"]+)"
| rex field=_raw "<Data Name=[\"]SourceIp[\"]>(?!<SourceIp>[^\"]+)"
| rex field=_raw "<Data Name=[\"]DestinationIp[\"]>(?!<DestinationIp>[^\"]+)"
| rex field=_raw "<Data Name=[\"]DestinationPort[\"]>(?!<DestinationPort>[^\"]+)"
| table _time SysmonEventCode Image SourceIp DestinationIp DestinationPort
```

Recommended Splunk Correlation

Alert on failures followed by success

```
index=main source="WinEventLog:Security" (EventCode=4625 OR EventCode=4624)
| stats count(eval(EventCode=4625)) AS failures
      count(eval(EventCode=4624)) AS successes
      earliest(_time) AS first_seen latest(_time) AS last_seen
      BY Source_Network_Address Account_Name
| where failures>=5 AND successes>=1
| convert ctime(first_seen) ctime(last_seen)
```

Recommended alert policy: trigger when five or more failures for the same source IP and account are followed by at least one successful logon within ten minutes. Review the successful event's Logon Type, target host, and account privileges before escalation.

Impact, Response, and Closure

Impact Assessment

- Observed impact: Successful network authentication to the localadmin account.
- Potential impact: Unauthorized system access, privilege abuse, credential reuse, and lateral movement.
- Not observed in supplied evidence: Persistence, destructive execution, privilege escalation, or data exfiltration.

Recommended Actions

- Reset or disable localadmin credentials and review membership in local privileged groups.
- Enable account lockout and enforce strong, unique passwords; use MFA where supported.
- Restrict SMB/RPC and remote administrative access to authorized management systems.
- Block or isolate suspicious source addresses during a real incident and preserve relevant logs.
- Deploy the Splunk correlation rule above and tune thresholds against normal administrative activity.
- Monitor Universal Forwarder health and validate XML field extraction for Sysmon data.

Evidence Register

Fig.	Evidence	Classification
1	Failed logins by IP	Primary evidence
2	Attacker IP / workstation field	Supporting evidence
3	Targeted account	Primary evidence
4	Broad 4624/4625 timeline	Timeline context
5	Initial Event ID 3 zero result	Diagnostic record
6	First combined attack timeline	Investigation context
7	Sysmon and authentication correlation view	Investigation context
8	Successful logons after failures	Primary evidence
9	Local Sysmon Event ID 3	Diagnostic record
10	Sysmon forwarding restored	Recovery verification
11	Event ID 3 network connection	Telemetry verification

Conclusion

Phase 09 is complete. The investigation reconstructed a successful credential attack against WIN11-01, documented the evidence and its limitations, restored Sysmon collection, and produced a repeatable Splunk detection approach. The SOC lab can now proceed to Phase 10.